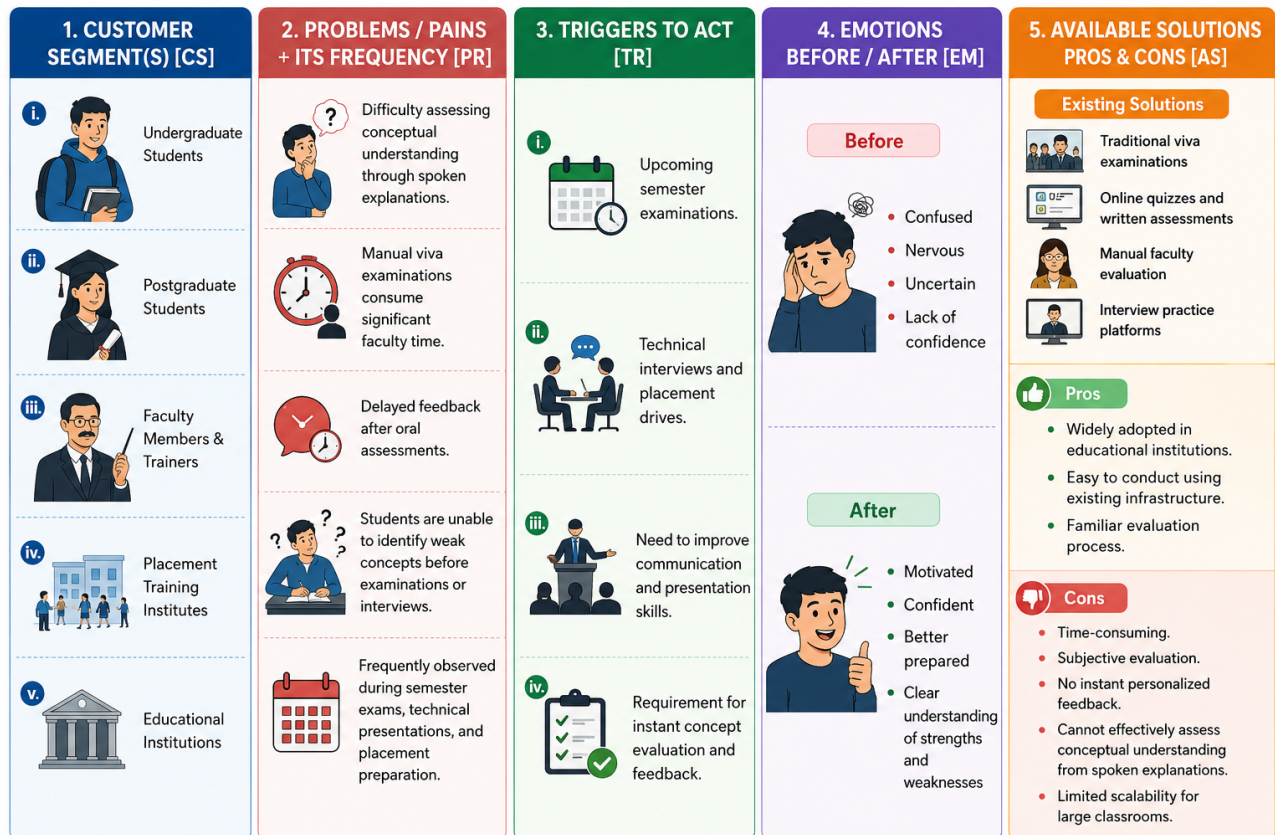


Problem Solution-Fit

Date	18 July 2026
Team ID	XXXXX
Project Name	Voice Based concept Understanding Analyser
Maximum Marks	5 Marks

Problem-Solution Fit :



6. CUSTOMER LIMITATIONS [CL]



Limited access to AI-based learning tools.



Lack of immediate personalized feedback.



Difficulty evaluating communication and conceptual clarity independently.



Limited opportunities to practice oral explanations before interviews.



Dependence on faculty availability for assessment.

7. BEHAVIOR + ITS INTENSITY [BE]



i. Practices concepts before examinations.



ii. Watches online tutorials to improve understanding.



iii. Frequently rehearses interview questions.



iv. Regularly seeks feedback from teachers or peers.



v. High usage before placements and semester examinations.

8. CHANNELS OF BEHAVIOR [CH]

ONLINE



i. Voice-Based Concept Understanding Analyser

ii. Learning Management Systems (LMS)

iii. YouTube Educational Channels

iv. NPTEL / SWAYAM Courses

v. GitHub Learning Resources

OFFLINE



• Classroom lectures

• Laboratory sessions

• Group discussions

• Technical workshops

• Mock interview sessions

9. PROBLEM ROOT / CAUSE [RC]



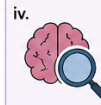
i. Traditional evaluation methods primarily assess written knowledge.



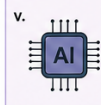
ii. Lack of automated systems for evaluating spoken conceptual understanding.



iii. Delayed and inconsistent feedback from manual assessments.



iv. Difficulty measuring semantic understanding rather than keyword matching.



v. Limited AI-enabled educational assessment platforms.

10. YOUR SOLUTION [SL]



Voice-Based Concept Understanding Analyser is an AI-powered educational assessment system that evaluates a learner's conceptual understanding through voice explanations. The application accepts recorded or uploaded audio, converts speech into text using the Whisper Speech Recognition model, and compares the explanation with reference content using Sentence-BERT semantic similarity. Based on the evaluation, the system generates similarity scores, grammar analysis, AI-based feedback, performance dashboards, downloadable PDF reports, and completion certificates. This enables students to improve conceptual clarity, communication skills, and interview readiness while reducing manual evaluation effort for educators.



Performance Dashboard



PDF Report



Completion Certificate